

# Heart Disease Prediction using Machine Learning

## Overview

This project focuses on predicting the presence of heart disease in patients based on various medical attributes. Several machine learning algorithms were implemented and compared to identify the most effective prediction model.

## Dataset

The dataset contains medical information such as:

- Age
- Sex
- Chest Pain Type
- Cholesterol
- fbs
- slope
- thalach
- target

## Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

## Machine Learning Models

The following algorithms were implemented:

- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest

## Project Workflow

1. Data Collection
2. Data Preprocessing
3. Exploratory Data Analysis (EDA)
4. Feature Selection

## 5. Final model

### **Results**

The models were evaluated using standard classification metrics such as:

- Accuracy
- Precision
- Recall
- F1 Score

### **About**

Heart-Disease-Prediction Machine Learning project for predicting heart disease risk using classification algorithms.